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Exercise - I Some Basic Concepts of Chemistry - Mole Concept

1. Number of moles of neutrons in 1.8 mL of water is :-
(1) 4.8×10^{23}
(2) 4.8×10^{22}
(3) 0.8
(4) 0.1
CMC225

2. 10 mL of gaseous hydrocarbon on complete combustion gives 20 mL of CO_2 and 30 mL of $H_2O(g)$. The hydrocarbon is :-
(1) C_4H_{10} (2) C_2H_6
(3) C_4H_8 (4) C_8H_{10}
CMC226

3. For the reaction A + 2B → C, 5 moles of A and 8 moles of B will produce :-
(1) 5 moles of C
(2) 4 moles of C
(3) 8 moles of C
(4) 13 moles of C
CMC227

4. Two elements X and Y (atomic mass of X = 75 and Y = 16) combine to give a compound having 76% of X. The formula of compound is :-
(1) XY (2) X_2Y
(3) X_2Y_2 (4) X_2Y_3
CMC228

5. 8 g H_2 & 32 g O_2 is allowed to react to form water then which of the following statement is correct ?
(1) O_2 is limiting reagent
(2) O_2 is reagent in excess
(3) H_2 is limiting reagent
(4) 40 g water is formed
CMC229

6. The maximum number of molecules are present in :-
(1) 15 L H_2 gas at STP
(2) 5 L of N_2 gas at STP
(3) 0.5 g H_2 gas
(4) 10 g of O_2 gas
CMC230

7. Equal masses of H_2, N_2 and CH_4 are taken in a container of volume V at temperature 27°C in identical conditions. The ratio of volumes of gases $H_2 : N_2 : CH_4$ would be :-
(1) 56 : 4 : 7
(2) 7 : 4 : 56
(3) 2 : 28 : 16
(4) 8 : 14 : 1
CMC231

8. Which of the following has the highest mass ?
(1) 1 g-atom of C
(2) $\frac{1}{2}$ mole of CH_4
(3) 10 mL of water
(4) 3.011×10^{23} atoms of oxygen
CMC232

9. At same temperature and pressure, 1 litre of a hydrocarbon weighs as much as one litre of CO_2. The molecular formula of hydrocarbon is :-
(1) C_3H_8
(2) C_2H_6
(3) C_2H_4
(4) C_3H_6
CMC233

10. Four one litre flasks are separately filled with the gases hydrogen, helium, oxygen and ozone at same room temperature and pressure in the different flasks. The ratio of total atoms would be :-
(1) 1 : 1 : 1 : 1
(2) 1 : 2 : 2 : 3
(3) 2 : 1 : 2 : 3
(4) 2 : 1 : 3 : 2
CMC234

11. In a compound A_xB_y :-
(1) Mole of A = mole of B = mole of A_xB_y
(2) eq of A = eq of B = eq of A_xB_y
(3) y × mole of A = y × mole of B = (x + y) mole of A_xB_y
(4) y × mole of A = y × mole of B
CMC235

12. The density of water is 1 g/mL. What is the volume occupied by 1 molecule of water in mL.

(1) 3×10^{-23} (2) 1×10^{-23}

(3) 2×10^{-23} (4) 4×10^{-23}

CMC236

13. The volume of a drop of water is 0.0018 mL then the number of water molecules present in two drops of water at room temperature is :-

(1) 12.046×10^{19} (2) 1.084×10^{18}

(3) 4.84×10^{17} (4) 6.023×10^{23}

CMC237

14. The law of multiple proportion is illustrated by the two compounds :-

(1) Sodium chloride and sodium bromide

(2) Ordinary water and heavy water

(3) Caustic soda and caustic potash

(4) Sulphur dioxide and sulphur trioxide

CMC238

15. The molecular mass of a gas is 45. Its density (in g/L) at STP is :-

(1) 22.4

(2) 11.2

(3) 5.7

(4) 2.0

CMC239

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Exercise-I ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	**3**	**2**	**2**	**4**	**1**	**1**	**1**	**1**	**1**	**3**	**2**	**1**	**1**	**4**	**4**

Exercise - II Some Basic Concepts of Chemistry - Mole Concept

1. The weight of 350 mL of a diatomic gas at 0°C and 2 atm pressure is 1 g. The weight of one atom is :-

(1) $\frac{16}{N_A}$

(2) $\frac{32}{N_A}$

(3) 16 N_A

(4) 32 N_A

CMC240

2. If the atomic weight of carbon is set at 24 amu, the value of the avogadro constant would be :-

(1) 6.022×10^{23}

(2) 12.044×10^{23}

(3) 3.011×10^{23}

(4) none of these

CMC241

3. Boron has two isotopes, B-10 and B-11. The average atomic mass of Boron is found to be 10.80 u. Calculate the percentage abundance of B^{10} isotope :-

(1) 80

(2) 20

(3) 25

(4) 75

CMC242

4. A compound on analysis gave the following results C = 54.54%, H = 9.09% and vapour density of compound = 88. Determine the molecular formula of the compound:-

(1) $C_8H_{16}O_4$

(2) $C_4H_{16}O_8$

(3) C_2H_4O

(4) CH_4O_2

CMC243

5. Mass of CO_2 produced on heating 20 g of 40% pure limestone :-

(1) 8 g

(2) 8.8 g

(3) 3.52 g

(4) None of these

CMC244

6. A compound contains 3.4% by weight of sulphur, the minimum molecular weight of compound is :-

(1) 941.176

(2) 944

(3) 945.27

(4) None

CMC245

7. If the weight of metal chloride is x g containing y g of metal, the equivalent weight of metal will be :-

(1) $\frac{x}{y} \times 35.5$

(2) $\frac{8(y-x)}{x}$

(3) $\frac{y}{x-y} \times 35.5$

(4) $E = \frac{8(x-y)}{y}$

CMC246

8. How many moles of magnesium phosphate $Mg_3(PO_4)_2$ will contain 0.25 mole of oxygen atoms ?

(1) 2.5×10^{-2}

(2) 0.02

(3) 3.125×10^{-2}

(4) 1.25×10^{-2}

CMC247

9. The number of H-atoms present in 5.6 g urea are

(1) 6.02×10^{23}

(2) 2.24×10^{23}

(3) 2.24×10^{22}

(4) 3.1×10^{22}

CMC248

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10. The isotopic abundance of C-12 and C-14 is 98% and 2% respectively. What would be the number of atoms of C-14 isotope in 12 g carbon sample
(1) 1.2×10^{22}
(2) 3.01×10^{23}
(3) 5.88×10^{23}
(4) 6.02×10^{23}

CMC249

11. The volume of air needed for complete combustion of 1 kg carbon at STP is
(1) 9333.3 litre
(2) 933.33 litre
(3) 93.33 litre
(4) 1866.67 litre

CMC250

12. The hydrated salt $Na_2CO_3.\ xH_2O$ undergoes 63% loss in mass on heating and becomes anhydrous. The value of x is :-
(1) 10
(2) 12
(3) 8
(4) 18

CMC251

13. 20 mL of methane is completely burnt using 50 mL of oxygen. The volume of the gas left after cooling to room temperature is :-
(1) 80 mL
(2) 40 mL
(3) 60 mL
(4) 30 mL

CMC252

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Exercise-II ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13
Answer	**1**	**2**	**2**	**1**	**3**	**1**	**3**	**3**	**2**	**1**	**1**	**1**	**4**

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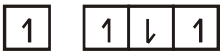
Exercise - I — Atomic Structure

1. Ionisation energy of He^+ is 19.6×10^{-18} J atom^{-1}. The energy of first stationary state (n =1) of Li^{+2} is :-
(1) 4.41×10^{-16} J atom^{-1}
(2) -4.41×10^{-17} J atom^{-1}
(3) -2.20×10^{-15} J atom^{-1}
(4) 8.82×10^{-17} J atom^{-1}
CAS291

2. If the radius of the first bohr orbit is X, the de Broglie wavelength of the electron in the third orbit is nearly:-
(1) $2\pi X$ (2) $6\pi X$
(3) 9X (4) X/3
CAS292

3. According to Bohr's theory, the angular momentum of an electron in 5th orbit is :-
(1) $25\frac{h}{\pi}$ (2) $1.0\frac{h}{\pi}$
(3) $10\frac{h}{\pi}$ (4) $2.5\frac{h}{\pi}$
CAS293

4. How many orbitals are possible for n = 3, l = 2 ?
(1) 1 (2) 3
(3) 5 (4) 7
CAS294

5. Which of the following is violating Hund's rule:-
(1) 
(2) 
(3) 
(4) All
CAS295

6. 0.5 g particle has 2×10^{-5} m uncertainty in position find the uncertainty in its velocity (m/s)
(1) 3.0×10^{33} (2) 5.27×10^{-27}
(3) 4×10^{-30} (4) 4×10^{-10}
CAS296

7. Which of the following has similar spectrum as that of Li^+ :-
(1) H (2) Na^+
(3) He (4) He^+
CAS297

8. Orbital angular momentum of d electron is:-
(1) $\sqrt{\frac{5}{2}}\frac{h}{\pi}$ (2) $\sqrt{6}\frac{h}{2\pi}$
(3) $\frac{h}{\sqrt{2}\pi}$ (4) $\frac{\sqrt{3}h}{2\pi}$
CAS298

9. Which has lowest wavelength :-
(1) Balmer series
(2) Brackett series
(3) Humphrey series
(4) Lyman series
CAS299

10. Which of the following ion has magnetic moment is 3.87 BM:-
(1) Ti^{3+} (2) Sc^+
(3) Ti^+ (4) Mn^{+5}
CAS300

11. Number of electrons in 2d orbital are :-
(1) 10 (2) 5
(3) 2 (4) zero
CAS301

12. Which of the following value of quantum number is not possible for 4f :-
(1) n = 4
(2) $\ell = 2$
(3) m = – 2 to +2
(4) all are possible
CAS302

13. Which of the following sets of quantum number is not possible
(1) n = 3, ℓ = +2, m_l = 0, m_s = +1/2
(2) n = 3, ℓ = 0, m_l = 0, m_s = –1/2
(3) n = 3, ℓ = 0, m_l = –1, m_s = +1/2
(4) n = 3, ℓ = 1, m_l = 0, m_s = + 1/2
CAS303

14. The number of radial nodes in 3s and 2p subshells are respectively

(1) 2 and 0 (2) 1 and 2

(3) 0 and 2 (4) 2 and 1

CAS304

15. How many electrons in an atom can have n = 3, l = 1, m = –1 and s = +½

(1) 1 (2) 2 (3) 4 (4) 6

CAS305

16. Given Rydberg constant R = 10^5 cm^{-1}. If electron jumps from M shell to K shell of H-atom, the frequency of the radiation emitted in cycle/s would be–

(1) $\frac{8}{9}\times10^5$ (2) $\frac{8}{3}\times10^{15}$

(3) $\frac{8}{3}\times10^{11}$ (4) $\frac{8}{9}\times10^{15}$

CAS306

17. Number of waves produced by an electron in one complete revolution in n^{th} orbit is :-

(1) n (2) n^2

(3) (n + 1) (4) (2n + 1)

CAS307

18. A dye absorbs a photon of wavelength λ and re-emits the same energy into two photons of wavelengths λ_1 and λ_2 respectively. The wavelength λ is related with λ_1 and λ_2 is :-

(1) $\lambda=\frac{\lambda_1+\lambda_2}{\lambda_1\lambda_2}$

(2) $\lambda=\frac{\lambda_1\lambda_2}{\lambda_1+\lambda_2}$

(3) $\lambda=\frac{\lambda_1^2\lambda_2^2}{\lambda_1+\lambda_2}$

(4) $\lambda=\frac{\lambda_1\lambda_2}{(\lambda_1+\lambda_2)^2}$

CAS308

Exercise-I ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	**2**	**2**	**4**	**3**	**4**	**2**	**3**	**2**	**4**	**3**	**4**	**2**	**3**	**1**	**1**
Question	16	17	18												
Answer	**2**	**1**	**2**												

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Exercise - II Atomic Structure

1. Energy of 1 mole of radio wave photons with a frequency of 909 kHz is :-

(1) 6.02×10^{-28}J (2) 3.62×10^{-4}J

(3) 1.00×10^{-4}J (4) 6.02×10^{-3}J

CAS309

2. Kinetic energy of an electron in the second Bohr orbit of a hydrogen atom is (a_0 is Bohr radius):-

(1) $\frac{h^2}{4\pi m a_0^2}$ (2) $\frac{h^2}{16\pi^2 m a_0^2}$

(3) $\frac{h^2}{32\pi^2 m a_0^2}$ (4) $\frac{h^2}{64\pi^2 m a_0^2}$

CAS310

3. In an atom 2K, 8L, 8M, 2N electrons are present. If m = 0, s = +1/2, then find no. of electrons :-

(1) 6 (2) 2

(3) 8 (4) 16

CAS311

4. Number of spectral line obtained when electron is provided with a photon of energy of 11 eV (for H atom)

(1) 1 (2) 2 (3) 0 (4) 6

CAS312

5. For which of the following sets of quantum numbers of an electron will have the highest energy?

	n	ℓ	m	s
(1)	3	2	1	–½
(2)	4	1	–1	+½
(3)	4	3	–1	+½
(4)	5	0	0	–½

CAS313

6. Calculate de Broglie wavelength of a neutron (mass, m= 1.6×10^{-27} kg) moving with kinetic energy of 0.04 eV.

(1) 146 Å (2) 14.6 Å

(3) 1460 Å (4) 1.46 Å

CAS314

7. An excited state of H atom emits a photon of wavelength λ and returns in the ground state, the principal quantum number of excited state is given by :-

(1) $\sqrt{\lambda R(\lambda R-1)}$ (2) $\sqrt{\frac{\lambda R}{(\lambda R-1)}}$

(3) $\sqrt{\lambda R}$ (4) $\sqrt{\frac{(\lambda R-1)}{\lambda R}}$

CAS315

Exercise-II ANSWER KEY

Question	1	2	3	4	5	6	7
Answer	**2**	**3**	**1**	**3**	**3**	**4**	**2**

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Exercise - I — Redox Reactions

1. In the reaction $SO_2 + H_2S \rightarrow 3S + 2H_2O$ the substance oxidised is :-
(1) S (2) SO_2 (3) H_2S (4) H_2O
CRR166

2. The oxidation number of phosphorous in $Ba(H_2PO_2)_2$ is :-
(1) +3 (2) +2 (3) +1 (4) –1
CRR167

3. The oxidation number of 'P' in $Mg_2P_2O_7$ is :-
(1) +3 (2) +2 (3) +5 (4) –3
CRR168

4. The number of electrons lost in the following change is :-
$Fe + H_2O \rightarrow Fe_3O_4 + H^+$
(1) 2 (2) 4 (3) 6 (4) 8
CRR169

5. Which of the following is a redox reaction :-
(1) $SO_2 + H_2O \longrightarrow H_2SO_3$
(2) $2CuSO_4 + 4KI \longrightarrow 2CuI + I_2 + 2K_2SO_4$
(3) $CuSO_4 + 4NH_3 \longrightarrow [Cu(NH_3)_4]\ SO_4$
(4) $C_{12}H_{22}O_{11} + H_2O \longrightarrow C_6H_{12}O_6 + C_6H_{12}O_6$
CRR170

6. For decolourisation of 1.5 moles of $KMnO_4$ the moles of H_2O_2 requires in acidic medium is :-
(1) $\frac{3}{2}$ (2) $\frac{9}{4}$ (3) $\frac{15}{4}$ (4) $\frac{21}{4}$
CRR171

7. In which change, transfer of five electrons takes place ?
(1) $CrO_4^{2-} \longrightarrow Cr^{+3}$
(2) $MnO_4^- \longrightarrow MnO_2$
(3) $Cr_2O_7^{2-} \longrightarrow 2Cr^{+3}$
(4) $MnO_4^- \longrightarrow Mn^{2+}$
CRR172

8. In the equation $NO_2^- + H_2O \rightarrow NO_3^- + 2H^+ + ne^-$, value of n is :-
(1) 1 (2) 2
(3) 3 (4) None of these
CRR173

9. In the reduction of dichromate by Fe(II), the number of electrons involved per chromium atom is :-
(1) 2 (2) 1 (3) 3 (4) 4
CRR174

10. The oxidation number of C in $K_2C_2O_4$ is
(1) +3 (2) +4 (3) 0 (4) +1
CRR175

11. $As_2S_3 + HNO_3 \rightarrow H_3AsO_4 + H_2SO_4 + NO$
the element which is oxidised in the reaction is/are
(1) As (2) S
(3) N (4) As and S
CRR176

12. Equivalent weight of FeS_2 in the half reaction, $FeS_2 \rightarrow Fe_2O_3 + SO_2$ is :-
(1) M/10 (2) M/11 (3) M/6 (4) M/1
CRR177

13. Equivalent weight of H_3PO_2 when it disproportionate into PH_3 and H_3PO_3 is :-
(1) M (2) M/2 (3) M/4 (4) 3M/4
CRR178

14. When BrO_3^- reacts with Br^- in acidic medium, Br_2 is liberated. The equivalent weight of Br_2 in this reaction is:-
(1) $\frac{5M}{8}$ (2) $\frac{5M}{3}$ (3) $\frac{3M}{5}$ (4) $\frac{4M}{6}$
CRR179

15. Hydrazine reacts with KIO_3 in presence of HCl as $N_2H_4 + IO_3^- + 2H^+ + Cl^- \rightarrow ICl + N_2 + 3H_2O$
The equivalent masses of N_2H_4 and KIO_3 respectively are :-
(1) 8 and 53.5
(2) 16 and 53.5
(3) 8 and 35.6
(4) 8 and 87
CRR180

Exercise-I — ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	3	3	4	2	3	4	2	3	1	4	2	4	3	1

Exercise - II
Redox Reactions

1. Which of the following reaction do not involve oxidation-reduction ?
(I) $2Cs + 2H_2O \rightarrow 2CsOH + H_2$
(II) $2CaI_2 \rightarrow 2Ca + 2I_2$
(III) $NH_4Br + KOH \rightarrow KBr + NH_3 + H_2O$
(IV) $4KCN + Fe(CN)_2 \rightarrow K_4[Fe(CN)_6]$
(1) I, II (2) I, III
(3) I, III, IV (4) III, IV

CRR181

2. In the reaction
$3Br_2+6CO_3^{2-} + 3H_2O \rightarrow 5Br^{\ominus} + BrO_3^{\ominus} + 6HCO_3^{\ominus}$
(1) Bromine is oxidised and carbonate is reduced
(2) Bromine is reduced and water is oxidised
(3) Bromine is neither reduced nor oxidised
(4) Bromine is both reduced and oxidised

CRR182

3. Which of the following is not a disproportionation reaction :-
(I) $NH_4NO_3 \rightarrow N_2O + H_2O$
(II) $P_4 \rightarrow PH_3 + H_2PO_2^-$
(III) $PCl_5 \rightarrow PCl_3 + Cl_2$
(IV) $IO_3^{\ominus} + I^{\ominus} \rightarrow I_2$
(1) I, II (2) I, III, IV
(3) II, IV (4) I, III

CRR183

4. In which reaction H_2O_2 acts as reducing agent?
(1) $Ag_2O + H_2O_2 \rightarrow 2Ag+H_2O + O_2$
(2) $2KI + H_2O_2 \rightarrow 2KOH + I_2$
(3) $PbS + 4H_2O_2 \rightarrow PbSO_4 + 4H_2O$
(4) $H_2O_2 + SO_2 \rightarrow H_2SO_4$

CRR184

5. What mass of HNO_3 is needed to convert 5g of iodine into iodic acid according to the reaction,
$I_2 + HNO_3 \rightarrow HIO_3 + NO_2 + H_2O$:-
(1) 12.4g (2) 24.8g
(3) 0.248 g (4) 49.6g

CRR185

6. How many grams of I_2 are present in a solution which requires 40 mL of 0.11 N $Na_2S_2O_3$ to react with it :-
$I_2 + S_2O_3^{2-} \rightarrow S_4O_6^{2-} + 2I^-$
(1) 12.7g (2) 0.558g
(3) 25.4 g (4) 11.4g

CRR186

7. In balancing the half reaction, $S_2O_3^{2-} \rightarrow S$, the number of electrons that must be added is :-
(1) 2 on the right
(2) 2 on the left
(3) 4 on the left
(4) 4 on the right

CRR187

8. In alkaline medium $KMnO_4$ reacts as follows :-
$2KMnO_4 + 2KOH \rightarrow 2K_2MnO_4 + H_2O + O_2$
Therefore its equivalent mass will be
(1) 31.6 (2) 52.7
(3) 79 (4) 158

CRR188

9. In which of the following the oxidation number of oxygen has been arranged in increasing order :-
(1) $OF_2 < KO_2 < BaO_2 < O_3$
(2) $BaO_2 < KO_2 < O_3 < OF_2$
(3) $BaO_2 < O_3 < OF_2 < KO_2$
(4) $KO_2 < OF_2 < O_3 < BaO_2$

CRR189

10. If it is known that in $Fe_{0.96}O$, Fe is present in +2 and +3 oxidation state. What is the mole fraction of Fe^{2+} in the compound?
(1) 12/25 (2) 25/12
(3) 1/12 (4) 11/12

CRR190

Exercise-II
ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	**4**	**4**	**2**	**1**	**1**	**2**	**3**	**4**	**2**	**4**

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Exercise - I Thermodynamics

1. A system absorb 20 kJ heat and does 10 kJ work then internal energy of system will be -
(1) Decreases by 10 kJ
(2) Increases by 10 kJ
(3) Increases by 30 kJ
(4) Decreases by 30 kJ

CTD343

2. Three moles of an ideal gas expanded spontaneously in to vacuum. Then which is correct?
(1) $W = 0, \Delta G = 0$ (2) $W = 0, \Delta G < 0$
(3) $W = 0, \Delta G > 0$ (4) $W \neq 0, \Delta G = 0$

CTD344

3. Which of the following is like state function (a) q + w, (b) q (c) w (d) heat in isobaric process (e) work in adiabatic process :-
(1) a, b, c (2) a, e (3) a, d, e (4) a, d

CTD345

4. Consider the following reaction :-

(a) $H^{+}_{(aq)} + Cl^{-}_{(aq)} \rightarrow HCl_{(\ell)}\ ; \Delta H = -X_1\ kJ\,mol^{-1}$

(b) $H_{2(g)} + Cl_{2(g)} \rightarrow 2HCl_{(\ell)}\ ; \Delta H = -X_2\ kJ\,mol^{-1}$

(c) $\frac{1}{2}H_{2(g)} + \frac{1}{2}Cl_{2(g)} \rightarrow HCl_{(\ell)}\ ; \Delta H = -X_3\ kJ\,mol^{-1}$

(d) $H_{(g)} + Cl_{(g)} \rightarrow HCl_{(\ell)}\ ; \Delta H = -X_4\ kJ\,mol^{-1}$

Enthalpy of formation of $HCl_{(\ell)}$ is :
(1) $-X_1$ kJ mol^{-1} (2) $-X_2$ kJ mol^{-1}
(3) $-X_3$ kJ mol^{-1} (4) $-X_4$ kJ mol^{-1}

CTD346

5. The following reactions are given :

$CH_{4(g)} + 2O_{2(g)} \rightarrow CO_{2(g)} + 2H_2O_{(g)}\ ; \Delta H = -x$ ...(1)

$C_{(s)} + O_{2(g)} \rightarrow CO_{2(g)}\ ; \Delta H = -y$ ...(2)

$H_{2(g)} + \frac{1}{2}O_{2(g)} \rightarrow H_2O_{(g)}\ ; \Delta H = -z$ ...(3)

Calculate the heat of formation of CH_4 ?
(1) x + y + z (2) y + 2z – x
(3) x – y – 2z (4) none of the above

CTD347

6. ΔU° of combustion of methane is –X kJ mol^{-1}
The value of ΔH° is (T = 25°C) :-
(1) –X – 596R (2) –X+596R
(3) X+596R (4) X–596R

CTD348

7. During adiabatic expansion of ideal gas, which is correct ?
(1) Temperature increases
(2) q = 0
(3) Temperature remain constant
(4) $\Delta E = 0$

CTD349

8. Enthalpy change for the reaction
$2N_{(g)} + 6H_{(g)} \rightarrow 2NH_{3(g)}$ is –900 kJ
The dissociation energy of N–H bond is
(1) +450 kJ (2) –450 kJ
(3) +150 kJ (4) –150kJ

CTD350

9. Which relation is incorrect :-
(1) $\Delta G = -T\,\Delta S_T$
(2) $\Delta G^\circ = -2.303\ RT \log K$
(3) $\Delta H = \Delta U + \Delta n_g RT$
(4) $w_{useful} = \Delta H$

CTD351

10. 2 mol of an ideal gas at 27°C expands isothermally and reversibly from a volume of 4L to 40 L. The work for this process (in kJ) is:-
(1) w = – 28.72 kJ (2) w = – 11.488 kJ
(3) w = – 5.736 kJ (4) w = – 4.988 kJ

CTD352

11. Match the column :-

	Column-I		Column-II
A	Adiabatic process	P	q = 0
B	Isothermal process	Q	$\Delta H = 0$
C	Isoenthalpic process	R	$\Delta T = 0$
D	Isoentropic process	S	$\Delta S = 0$

(1) A→ P, B→ S, C→ Q, D→ R
(2) A→ Q, B→ P, C→ S, D→ R
(3) A→ P, B→ R, C→ Q, D→ S
(4) A→ P, B→ R, C→ S, D→ Q

CTD353

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12. $CH_4 + \frac{1}{2}O_2 \rightarrow CH_3OH$; $\Delta H = -ve$

If enthalpy of combustion of CH_4 and CH_3OH is -x & -y respectively which relation is correct?

(1) x > y (2) x < y

(3) x = y (4) x ≥ y

CTD354

13. The standard enthalpies of formation of CO, NO_2 and CO_2 are –110.5 kJ mol^{-1}, –33.2 kJ mol^{-1} and –393.5 kJ mol^{-1} respectively. The standard enthalpy of the reaction is :-

$4CO(g) + 2NO_2(g) \rightarrow 4CO_2(g) + N_2(g)$ is

(1) – 1065.6 kJ (2) – 200 kJ

(3) – 700 kJ (4) 850 kJ

CTD355

14. 18 g of water is taken to prepare the tea. Find out the internal energy of vaporisation at 100°C. (Δ_{vap} H for water at 373 K is 40.66 kJ mol_{-1})

(1) 37.56 kJ mol^{-1} (2) –37.56 kJ mol^{-1}

(3) 43.76 kJ mol^{-1} (4) –43.76 kJ mol^{-1}

CTD356

15. Which among the following is an extensive property of the system ?

(1) temperature (2) volume

(3) refractive index (4) viscosity

CTD357

16. $Na(s) \rightarrow Na(g)$, the heat of reaction is called as

(1) Heat of vaporisation

(2) Heat of atomisation

(3) Heat of sublimation

(4) Both (2) and (3)

CTD358

17. Which of the following condition regarding a chemical process ensures spontaneity at all temp. ?

(1) $\Delta H < 0, \Delta S > 0$

(2) $\Delta H > 0, \Delta S < 0$

(3) $\Delta H < 0, \Delta S < 0$

(4) $\Delta H > 0, \Delta S > 0$

CTD359

18. A reaction occur spontaneously. If

(1) $T\Delta S = \Delta H$ and both ΔH and ΔS are positive

(2) $T\Delta S > \Delta H$ and both ΔH and ΔS are positive

(3) $T\Delta S > \Delta H$ and both ΔH and ΔS are negative

(4) $T\Delta S = \Delta H$ and ΔH is positive and ΔS is negative

CTD360

19. 2 moles of an ideal gas is expanded isothermally and reversibly from 1 L to 10 L. Find the enthalpy change in kJ mol^{-1}.

(1) 0 (2) 11.7

(3) –11.7 (4) 25

CTD361

20. Which of the following endothermic process is/are spontaneous ?

(1) Melting of ice

(2) Evaporation of water

(3) Heat of combustion

(4) Both (1) and (2)

CTD362

Exercise-I ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	**2**	**2**	**3**	**3**	**3**	**1**	**2**	**3**	**4**	**2**	**3**	**1**	**1**	**1**	**2**
Question	16	17	18	19	20										
Answer	**4**	**1**	**2**	**1**	**4**										

Exercise - II Thermodynamics

1. Match the following in List-I with List-II and select the correct option :-

	List-I		List-II
a	$K_P = Q$	i	Always nonspontaneous
b	$T > \Delta H/\Delta S$	ii	Isothermal process
c	ΔH = +ve, ΔS = –ve	iii	Equilibrium
d	q = –w	iv	Spontaneous and endothermic

(1) a → (iii) b → (iv) , c → (i), d→ (ii)

(2) a → (i) b → (iii) , c → (ii), d→ (iv)

(3) a → (iii) b → (i) , c → (iv), d→ (ii)

(4) a → (iii) b → (ii) , c → (i), d→ (iv)

CTD363

2. Assume each reaction is carried in open container. For which reaction $\Delta H > \Delta E$?

(1) $H_{2(g)} + I_{2(g)} \rightarrow 2HI_{(g)}$

(2) $N_{2(g)} + 3H_{2(g)} \rightarrow 2NH_{3(g)}$

(3) $PCl_{5(g)} \rightarrow PCl_{3(g)} + Cl_{2(g)}$

(4) $SO_{2(g)} + \frac{1}{2} O_{2(g)} \rightarrow SO_{3(g)}$

CTD364

3. Given that Bond energies of H_2 and N_2 are 400 kJ mol^{-1} and 240 kJ mol^{-1} respectively and ΔH_f of NH_3 is –120 kJ mol^{-1}. Calculate the bond energy of N–H bond :-

(1) 300 kJ mol^{-1}

(2) 250 kJ mol^{-1}

(3) 410 kJ mol^{-1}

(4) 280 kJ mol^{-1}

CTD365

4. The equilibrium constant for a reaction is 100 what will be the value of $\Delta G°$?

R = 8.314 JK^{-1} mol^{-1}, T = 300 K :-

(1) –11488 kJ (2) –11.488 kJ

(3) –12 kJ (4) –12000 kJ

CTD366

5. For the reaction :-

$N_2O_{4(g)} \rightleftharpoons 2NO_{2(g)}$

ΔU = 2.0 kCal, ΔS = 50 Cal K^{-1} at 300 K calculate ΔG ?

(1) +12.4 kCal (2) –12.4 kCal

(3) –6.4 kCal (4) +6.4 kCal

CTD367

6. Consider the following process :-

$A \rightarrow 2B$; ΔH = + 150 kJ

$3B \rightarrow 2C + D$; ΔH = – 125 kJ

$G + A \rightarrow 2D$; ΔH = + 350 kJ

For $B + D \rightarrow G + 2C$; ΔH will be

(1) +525 kJ (2) +325 kJ

(3) –175 kJ (4) –325 kJ

CTD368

7. Match the column :-

A	$C_{(s,graphite)} + O_{2(g)} \rightarrow CO_{2(g)}$	P	$\Delta H^o_{Combustion}$
B	$CO_{(g)} + \frac{1}{2} O_{2(g)} \rightarrow CO_{2(g)}$	Q	$\Delta H^o_{sublimation}$
C	$CH_{4(g)} \rightarrow C_{(g)} + 4H_{(g)}$	R	$\Delta H^o_{formation}$
D	$C_{(s)} \rightarrow C_{(g)}$	S	$\Delta H^o_{atomisation}$

(1) A→ R, B→ S, C→ P, D→ Q

(2) A→ R, B→ P, C→ Q, D→ S

(3) A→ P, B→ S, C→ Q, D→ R

(4) A→ R, B→ P, C→ S, D→ Q

CTD369

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8. Match the column :-

	Sign of ΔH & ΔS respectively		Nature of reaction
A	– & –	P	Spontaneous only at low temperature
B	– & +	Q	Spontaneous only at high temperature
C	+ & +	R	Spontaneous at all temperature
D	+ & –	S	Nonspontaneous at all temperature

(1) A → P, B→ R, C→ Q, D→ S
(2) A → R, B→ P, C→ Q, D→ S
(3) A → Q, B→ R, C→ P, D→ S
(4) A → P, B→ Q, C→ R, D→ S

CTD370

9. The heat of neutralization of LiOH and H_2SO_4 at 25°C is –69.6 kJ mol^{-1}. The heat of ionisation of LiOH will be nearly :-

(1) 22.5 kJ mol^{-1} (2) 90 kJ mol^{-1}
(3) 45 kJ mol^{-1} (4) 33.6 kJ mol^{-1}

CTD371

10. If the ΔG° of a cell reaction

$AgCl + e^- \rightarrow Ag + Cl^-$ is –21.20 kJ

the standard emf of cell is

(1) 0.329V (2) 0.220V
(3) –0.220V (4) –0.110V

CTD372

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Exercise-II ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	1	3	4	2	2	4	4	1	1	2

Exercise - I — Chemical Equilibrium

1. In a system $P_{(s)} \rightleftharpoons 2Q_{(g)} + R_{(g)}$, at equilibrium if concentration of 'Q' is doubled then how many times the concentration of R at equilibrium will become :-
(1) Double of its original concentration
(2) $\frac{1}{4}$ of its original concentration
(3) $\frac{1}{2}$ of its original concentration
(4) 4 times of its original concentration
CCE189

2. 1 mol of $XY_{(g)}$ and 0.2 mol of $Y_{(g)}$ are mixed in 1L vessel. At equilibrium 0.6 mol of $Y_{(g)}$ is present. The value of K for the reaction :-
$XY(g) \rightleftharpoons X(g) + Y(g)$ is
(1) 0.04 (2) 0.06 (3) 0.36 (4) 0.4
CCE190

3. For the reaction :-
$X_{2(g)} + Y_{2(g)} + Z_{2(g)} \rightleftharpoons 2XYZ_{(g)}$
At equilibrium concentration of X_2 = 3M, Y_2 = 6M, Z_2 = 9M, XYZ = 6M. If the reaction takes place in a sealed vessel at 527°C, then the value of K_C will be :
(1) $\frac{27}{2}$ (2) $\frac{2}{9}$ (3) 36 (4) 24
CCE191

4. The equilibrium constant K_C for the following reaction at 842°C is 7.90×10^{-3}. What is K_P at same temperature ?
$$\frac{1}{2}F_{2(g)} \rightleftharpoons F_{(g)}$$
(1) 8.64×10^{-5} (2) 8.26×10^{-4}
(3) 7.90×10^{-2} (4) 7.56×10^{-2}
CCE192

5. $NH_4HS_{(s)} \rightleftharpoons NH_{3(g)} + H_2S_{(g)}$
The equilibrium pressure at 25°C is 0.660 atm What is K_p for reaction ?
$NH_4NH_{(s)} \rightleftharpoons NH_{3(g)} + H_2S_{(g)}$
(1) 0.109 (2) 0.218
(3) 1.89 (4) 2.18
CCE193

6. For the reaction $CO_{(g)} + Cl_{2(g)} \rightleftharpoons COCl_{2(g)}$ the value of $\frac{K_C}{K_P}$ is equal to :-
(1) $\sqrt{RT}$ (2) RT (3) $\frac{1}{RT}$ (4) 1
CCE194

7. The equilibrium constant (K_c) for the reaction
$$2HCl_{(g)} \rightleftharpoons H_{2(g)} + Cl_{2(g)}$$
is 2×10^{-2} at 25°C. What is the equilibrium constant for the reaction
$H_{2(g)} + Cl_{2(g)} \rightleftharpoons 2HCl_{(g)}$:-
(1) 2×10^{-2} (2) 50
(3) 5×10^{2} (4) None of these
CCE195

8. If K_1 and K_2 are the equilibrium constant for the two reactions
$XeF_{6(g)} + H_2O_{(g)} \rightleftharpoons XeOF_{4(g)} + 2HF_{(g)}$; K_1
$XeO_{4(g)} + XeF_{6(g)} \rightleftharpoons XeOF_{4(g)} + XeO_3F_{2(g)}$; K_2
The equilibrium constant for the reaction
$XeO_{4(g)} + 2HF_{(g)} \rightleftharpoons XeO_3F_{2(g)} + H_2O_{(g)}$ is
(1) $K_1 K_2$ (2) K_1/K_2^2
(3) K_2/K_1 (4) K_1/K_2
CCE196

9. At 25°C the equilibrium constant K_1 and K_2 of two reactions are
$2NH_3 \rightleftharpoons N_2 + 3H_2$; K_1
$\frac{1}{2}N_2 + \frac{3}{2}H_2 \rightleftharpoons NH_3$; K_2
The relation between K_1 and K_2 is :-
(1) $K_1 = K_2$ (2) $K_2 = \frac{1}{K_1^2}$
(3) $K_1 = \frac{1}{K_2^2}$ (4) $K_1 = \frac{1}{K_2}$
CCE197

10. For the reaction $2SO_3(g) \rightleftharpoons 2SO_{2(g)} + O_{2(g)}$ the value of $\frac{K_C}{K_p}$ will be :-
(1) $(RT)^1$ (2) $(RT)^{-1}$ (3) $\sqrt{RT}$ (4) $(RT)^{-2}$
CCE198

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11. For the reaction $2NO_{2(g)} + \frac{1}{2}O_{2(g)} \rightleftharpoons N_2O_{5(g)}$

if the equilibrium constant is K_p, then the equilibrium constant for the reaction

$2N_2O_{5(g)} \rightleftharpoons 4NO_{2(g)} + O_{2(g)}$ would be :-

(1) K_P^2 (2) $\frac{2}{K_p}$ (3) $\frac{1}{K_p^2}$ (4) $\frac{1}{\sqrt{K_p}}$

CCE199

12. In the reaction $X_{(g)} + Y_{(g)} \rightleftharpoons 2Z_{(g)}$

2 mol of X, 1 mol of Y and 1 mol of Z are placed in a 10 litre vessel and allowed to reach equilibrium. If final concentration of Z is 0.2M, then K_C for the given reaction is :-

(1) 1.6 (2) 80/3

(3) 16/3 (4) None of these

CCE200

13. For the reversible reaction

$PCl_{3(g)} + Cl_{2(g)} \rightleftharpoons PCl_{5(g)}$ + heat

The equilibrium will shift in forward direction:-

(1) By increasing concentration of $PCl_{5(g)}$

(2) By decreasing pressure

(3) By decreasing concentration of $PCl_{3(g)}$ and $Cl_{2(g)}$

(4) By increasing pressure and decreasing temp.

CCE201

14. For the reaction

$CO(g) + H_2O(g) \rightleftharpoons CO_2(g) + H_2(g)$

at a given temperature the equilibrium amount of $CO_2(g)$ can be increased by

(1) adding suitable catalyst

(2) adding inert gas

(3) decreasing the volume of container

(4) increasing the amount of CO(g)

CCE202

15. $NH_2COONH_4(s) \rightleftharpoons 2NH_3(g) + CO_2(g)$

If equilibrium pressure is 3 atm for the above reaction, K_P will be

(1) 4 (2) 27

(3) 4/27 (4) 1/27

CCE203

16. For the reaction $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$, the equilibrium constant will change with

(1) total pressure

(2) catalyst

(3) concentration of H_2 and I_2

(4) temperature

CCE204

17. Increase in the pressure for the following equilibrium

$H_2O(\ell) \rightleftharpoons H_2O(g)$ result in the

I : Formation of more $H_2O(\ell)$

II : Formation of more $H_2O(g)$

III : Increase in b.p. of $H_2O(\ell)$

IV : Decrease in b.p. of $H_2O(\ell)$

Hence, correct statements are–

(1) I, II (2) II, III (3) I, III (4) I, IV

CCE205

Exercise-I ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	**2**	**4**	**2**	**4**	**1**	**2**	**2**	**3**	**3**	**2**	**3**	**3**	**4**	**4**	**1**
Question	16	17													
Answer	**4**	**3**													

Exercise - II — Chemical Equilibrium

1. In which of the following reaction is almost completed :-
(1) $K_{eq} = 10$ (2) $K_{eq} = 1$
(3) $K_{eq} = 10^5$ (4) $K_{eq} = 10^{-2}$
CCE206

2. If the concentration of Y^- ion in the reaction
$$XY_{3(s)} \rightleftharpoons X^{+3}_{(aq)} + 3Y^-_{(aq)} \text{ is}$$
decreased $\frac{1}{2}$ times, then how many times the equilibrium concentration of X^{+3} will increase:
(1) 2 times (2) 4 times
(3) 8 times (4) 16 times
CCE207

3. A gaseous mixture was prepared by taking 3 mol of H_2 and 1 mol of CO. If the total pressure of the mixture was found 2 atmosphere then partial pressure of hydrogen (H_2) in the mixture is :-
(1) $\frac{3}{2}$ (2) $\frac{1}{2}$ (3) 1 (4) 2
CCE208

4. The value of K_c at 900 K temperature for following reaction is 5.
$N_{2(g)} + O_{2(g)} \rightleftharpoons 2NO_{(g)}$ if equilibrium concentration of N_2 and O_2 is 0.5M then value of K_P is :-
(1) 0.02 (2) 0.2 (3) 5 (4) $\frac{5}{RT}$
CCE209

5. Consider equilibrium constant for the reaction:
$2CO_{(g)} + 2H_2O_{(g)} \rightleftharpoons 2CO_{2(g)} + 2H_{2(g)}$ eq. const = K_1
$CH_{4(g)} + H_2O_{(g)} \rightleftharpoons CO_{(g)} + 3H_{2(g)}$; eq. const = K_2
$CH_{4(g)} + 2H_2O_{(g)} \rightleftharpoons CO_{2(g)} + 4H_{2(g)}$; eq. const = K_3
Which of the following relation is correct ?
(1) $K_3 = \frac{K_1}{K_2}$ (2) $K_3 = \frac{K_1^2}{K_2^2}$
(3) $K_3 = K_1K_2$ (4) $K_3 = \sqrt{K_1} \cdot K_2$
CCE210

6. At certain temperature compound $AB_{2(g)}$ dissociates according to the reaction
$$AB_{2(g)} \rightleftharpoons AB_{(g)} + B_{(g)}$$
with degree of dissociation α which is small compared to unity. The expression of K_P in terms of α and initial pressure P is :-
(1) $P\alpha^3$ (2) $P\alpha^2$
(3) $\frac{\alpha^3}{P}$ (4) $\frac{\alpha^2}{P}$
CCE211

7. Naphthalene, a white solid used to make moth balls has a vapour pressure of 0.10 mm Hg at 27°C. Hence K_p (in atm) and K_c for the equilibrium are:
$C_{10}H_8$ (s) $\rightleftharpoons$ $C_{10}H_8$ (g)
(1) 0.10, 0.10
(2) 0.10, 4.1×10^{-3}
(3) 1.32×10^{-4}, 5.34×10^{-6}
(4) 5.36×10^{-6}, 1.32×10^{-4}
CCE212

8. At a given temperature equilibrium is attained when 50% of each reactant is converted into the products
A(g) + B(g) $\rightleftharpoons$ C(g) + D(g)
If in other experiment initial amount of B(g) is doubled, percentage of B converted into products will be–
(1) 100%
(2) 50%
(3) 66.67%
(4) 33.33%
CCE213

Exercise-II — ANSWER KEY

Question	1	2	3	4	5	6	7	8
Answer	**3**	**3**	**1**	**3**	**4**	**2**	**3**	**4**

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Exercise - I — Ionic Equilibrium

1. What is the K_b of monoacidic weak base if a 0.05 M solution is 2.5% ionised ?
(1) 7.8×10^{-8} (2) 1.6×10^{-6}
(3) 3.1×10^{-5} (4) 1.2×10^{-3}
CIE342

2. For a sparingly soluble salt A_pB_q, the relation of its solubility product (L_s) with its solubility (S) is :-
(1) $L_S = S^{P+q}.\ P^P.\ q^q$ (2) $L_S = S^{P+q}\ P^q.\ q^P$
(3) $L_S = S^{Pq}.\ P^P.\ q^q$ (4) $L_S = S^{Pq}.\ (Pq)^{P+q}$
CIE343

3. Find the pH of a solution prepared by mixing 25 ml of a 0.5M solution of HCl, 10 ml of a 0.5 M solution of NaOH and 15 ml of water :-
(1) 4 (2) 3
(3) 0.82 (4) 0.10
CIE344

4. Calculate the pH of a 10^{-3}M solution of $Ba(OH)_2$ if it undergoes complete ionisation. ($Kw = 1 \times 10^{-14}$) :-
(1) 12.30 (2) 11.30
(3) 10.00 (4) 9.00
CIE345

5. For poly basic acid, the dissociation constant have a different values for each step.

$H_3A \rightleftharpoons H^+ + H_2A^-$; $K_{eq} = K_{a_1}$

$H_2A^- \rightleftharpoons H^+ + HA^{2-}$; $K_{eq} = K_{a_2}$

$HA^{2-} \rightleftharpoons H^+ + A^{3-}$; $K_{eq} = K_{a_3}$

What is the observed trend of dissociation constant in successive stages ?
(1) $K_{a_1} > K_{a_2} > K_{a_3}$ (2) $K_{a_1} = K_{a_2} = K_{a_3}$
(3) $K_{a_1} < K_{a_2} < K_{a_3}$ (4) $K_{a_1} = K_{a_2} + K_{a_3}$
CIE346

6. NH_4CN is a salt of weak acid HCN ($K_a = 6.2\times10^{-10}$) and a weak base NH_4OH ($K_b = 1.8 \times 10^{-5}$) A molar solution of NH_4CN will be :-
(1) Neutral (2) Strongly acidic
(3) Strongly basic (4) Weakly basic
CIE347

7. The solubility product of $BaCl_2$ is 4×10^{-9}. What will be its solubility in mol/L :-
(1) 4×10^{-3} (2) 3.2×10^{-9}
(3) 10^{-3} (4) 1×10^{-9}
CIE348

8. Which of the following pairs can be used to form a buffer solution?
(1) H_2SO_4 and Na_2SO_4
(2) NaOCl and NaCl
(3) NH_3 and NH_4Cl
(4) NaH_2PO_4 and NaCl
CIE349

9. The number of OH^- ion in 1mL of solution having pH = 4 is
(1) 10^{-4} (2) 10^{-10}
(3) 6.02×10^{10} (4) 6.02×10^{13}
CIE350

Exercise-I — ANSWER KEY

Question	1	2	3	4	5	6	7	8	9
Answer	**3**	**1**	**3**	**2**	**1**	**4**	**3**	**3**	**3**

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Exercise - II — Ionic Equilibrium

1. Solubility of calcium phosphate (molecular mass, M) in water is w g per 100 ml at 25°C, its solubility product at 25°C will be approximately :-

(1) $10^9\left(\frac{w}{M}\right)^5$ (2) $10^7\left(\frac{w}{M}\right)^5$

(3) $10^5\left(\frac{w}{M}\right)^5$ (4) $10^3\left(\frac{w}{M}\right)^5$

CIE351

2. The K_{sp} of Ag_2CrO_4 is 1.1×10^{-12} at 298 K, solubility (mol/L) of Ag_2CrO_4 in 0.1M $AgNO_3$ solution is :-

(1) 1.1×10^{-11} (2) 1.1×10^{-10}

(3) 1.1×10^{-12} (4) 1.1×10^{-9}

CIE352

3. A certain weak acid has a dissociation constant 1.0×10^{-4}, the equilibrium constant for its reaction with strong base is :-

(1) 1.0×10^{-4} (2) 1.0×10^{-10}

(3) 1.0×10^{10} (4) 1.0×10^{14}

CIE353

4. A certain buffer solution contains equal concentration of X^- and HX. The K_b for X^- is 10^{-10} the pH of the buffer is :-

(1) 4 (2) 9 (3) 10 (4) 7

CIE354

5. 0.01 mole of sodium hydroxide is added to 10 litres of water, How much will the pH of water change?

(1) 4 (2) 7 (3) 5 (4) 8

CIE355

6. Calculate the pH of a 10^{-5} M HCl solution if 1ml of it is diluted to 1000 ml, Kw = 1×10^{-14} :-

(1) 7.98 (2) 6.98

(3) 7.0 (4) 5

CIE356

7. Calculate the degree of hydrolysis of the 0.01 M solution of salt KF [$(K_a)_{HF} = 6.6 \times 10^{-4}$]

(1) 3.87×10^{-6} (2) 3.87×10^{-5}

(3) 3.87×10^{-2} (4) None of these

CIE357

8. Dissociation constant of CH_3COOH and NH_4OH in aqueous solution are 10^{-5} if pH of a CH_3COOH solution is 3, what will be the pH of NH_4OH solution having same concentration as that of CH_3COOH solution ?

(1) 3.0 (2) 4.0 (3) 10.0 (4) 11.0

CIE358

9. What is the concentration of SO_4^{-2} required start the precipitation of $BaSO_4$ in a solution containing 1×10^{-4} mol L^{-1} of Ba^{2+}

(K_{sp} of $BaSO_4 = 4 \times 10^{-10}$) :-

(1) 4×10^{-10} M (2) 2×10^{-10} M

(3) 4×10^{-6} M (4) 2×10^{-3} M

CIE359

10. The conjugate base of strong acid in the reaction $CH_3COOH + HCl \rightleftharpoons Cl^- + CH_3COOH_2^+$ will be :-

(1) HCl (2) Cl^-

(3) CH_3COOH (4) $CH_3COOH_2^+$

CIE360

11. What is $\left[NH_4^+\right]$ in a solution that is 0.02 M NH_3 and 0.01 M KOH? [K_b (NH_3) = 1.8×10^{-5}]

(1) 3.6×10^{-5} M (2) 1.8×10^{-5} M

(3) 0.9×10^{-5} M (4) 7.2×10^{-5} M

CIE361

12. The pH of a solution obtained by mixing 100 mL of 0.2 M CH_3COOH with 100 mL of 0.2 M NaOH would be :

(pK_a for CH_3COOH = 4.74)

(1) 4.74 (2) 8.87 (3) 9.10 (4) 8.57

CIE362

Exercise-II — ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12
Answer	**2**	**2**	**3**	**1**	**1**	**2**	**2**	**4**	**3**	**3**	**1**	**2**

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Exercise - I — Electrochemistry

1. A hypothetical electrochemical cell is shown below

$Pt(s)|A_2(1atm)|A^-(xM)||B^+(yM)|B(s)$

The emf measured is 0.30V. The cell reaction is :-

(1) $A_2 + 2B^+ \rightarrow 2A^- + 2B$

(2) $A_2 + 2e^- \rightarrow 2A^-$; $2B^+ + 2e^- \rightarrow 2B$

(3) The cell reaction cannot be predicted

(4) $2A^- + 2B^+ \rightarrow A_2 + 2B$

CEL200

2. If $E^{\circ}_{A^{+2}/A} = -0.30$ V and $E^{\circ}_{A^{+3}/A^{+2}} = 0.40$ V, the standard Emf of the reaction : $A + 2A^{+3} \rightarrow 3A^{+2}$ will be :-

(1) 0.30 V (2) 0.40 V

(3) 0.70 V (4) 0.10 V

CEL201

3. Given $Fe^{+2} + 2e^- \rightarrow Fe(s)$ $E^\circ = -0.447$V

$Fe^{+2} \rightarrow Fe^{+3} + e^-$ $E^\circ = -0.771$V

Find E° for the reaction $Fe^{+3} + 3e^- \rightarrow Fe(s)$

(1) 0.34 V (2) –0.041V

(3) +0.39 V (4) –0.47 V

CEL202

4. An increase in molar conductance of a strong electrolyte with dilution is mainly due to :-

(1) Increase in number of ions

(2) Increase in ionic mobility of ions

(3) 100% ionisation of electrolyte at normal dilution

(4) Increase in both i.e. number of ions and ionic mobility of ions.

CEL203

5. Which relation of emf of an electrochemical cell is correct :-

(1) emf of cell = oxidation potential of anode – reduction potential of cathode

(2) emf of cell = oxidation potential of anode + reduction potential of cathode

(3) emf of cell = reduction potential of anode + reduction potential of cathode

(4) All of these

CEL204

6. If the E°_{cell} for a given reaction has a positive value, then which of the following gives the correct relationship for the values of ΔG° and K_{eq}:-

(1) $\Delta G^\circ < 0$; $K_{eq} < 1$ (2) $\Delta G^\circ > 0$; $K_{eq} > 1$

(3) $\Delta G^\circ > 0$; $K_{eq} < 1$ (4) $\Delta G^\circ < 0$; $K_{eq} > 1$

CEL205

7. Standard electrode potential of three metals A, B and C are –1.5V, 0.8V and –2.7V respectively. The reducing power of these metals will be :-

(1) B > C > A (2) B > A > C

(3) C > A > B (4) A > B > C

CEL206

8. The weight of copper (At wt = 63.5) displaced by a quantity of electricity which displaces 5600 ml of O_2 at STP will be :-

(1) 63.5 g (2) 31.75 g

(3) 15.875 g (4) 127 g

CEL207

9. When 0.1 mole $Cr_2O_7^{-2}$ is reduced then quantity of electricity required to reduced $Cr_2O_7^{-2}$ to Cr^{+3} completely is :-

(1) 9650 C (2) 96500 C

(3) 57900 C (4) 54900 C

CEL208

10. A device that converts energy of combustion of fuel like hydrogen and methane directly into electrical energy is known as :-

(1) Electrolytic cell (2) Dynamo

(3) Ni-Cd cell (4) Fuel cell

CEL209

11. The pressure of H_2 required (in bar) to make the potential of H_2-electrode zero in pure water at 25°C is :-

(1) 10^{-14} (2) 10^{-7} (3) 10^{+14} (4) 10^{+7}

CEL210

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12. The number of electrons delivered at the cathode during electrolysis by a current of 2 ampere in 120 seconds is
(charge on electron = 1.6×10^{-19} C) :-
(1) 3.75×10^{20} (2) 7.48×10^{20}
(3) 1.5×10^{21} (4) 3×10^{20}

CEL211

13. The molar conductivity of a 0.1 mol/dm^3 solution of KNO_3 with electrolytic conductivity of 4×10^{-3} Scm^{-1} at 298 K is :-
(1) 11.52 Scm^2 mol^{-1} (2) 20 Scm^2 mol^{-1}
(3) 40 Scm^2 mol^{-1} (4) 13.48 Scm^2 mol^{-1}

CEL212

14. The conductance is the :-
(1) Reciprocal of specific resistance
(2) Reciprocal of resistance
(3) Reciprocal of current
(4) Reciprocal of concentration

CEL213

15. Zn can displace :-
(1) Mg^{+2} from its aqueous solution
(2) Cu^{+2} from its aqueous solution
(3) Na^{+} from its aqueous solution
(4) Al^{+3} from its aqueous solution

CEL214

16. Which of the following is incorrect regarding salt bridge solution ?
(1) Solution must be of strong electrolyte
(2) Solution should be inert towards both electrodes
(3) Size of cation and anions of salt should be much different
(4) Salt bridge solution is prepared in gelatin or agar-agar to make it semi solid.

CEL215

17. The efficiency of fuel cell is given by :-
(1) $\frac{\Delta G}{\Delta S}$ (2) $\frac{\Delta G}{\Delta H}$
(3) $\frac{\Delta S}{\Delta G}$ (4) $\frac{\Delta H}{\Delta G}$

CEL216

18. Calculate E_{cell} of the reaction
$Mg(s) + 2Ag^{+} \rightarrow Mg^{+2} + 2Ag(s)$
(0.0001M) (0.100M)
If $E^0_{cell} = 3.17$ V
(1) –2.96 V (2) +2.96V
(3) 3.38 V (4) – 3.38 V

CEL217

19. Pt $|H_2(1atm)|H^{+}$ (0.001M)$||H^{+}$ (0.1M)$|H_2$ (1atm)|Pt
what will be the value of E_{cell} for this cell
(1) 0.1182 V
(2) – 0.1182 V
(3) 0.0591 V
(4) – 0.0591 V

CEL218

20. If $E^0_{Au^{+}/Au} = 1.69V$ and $E^0_{Au^{3+}/Au} = 1.40V$ then the value of $E^0_{Au^{+3}/Au^{+}}$ will be
(1) 0.19V (2) 2.945 V
(3) 1.255 V (4) 1.19 V

CEL219

Exercise-I ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	**4**	**3**	**2**	**2**	**2**	**4**	**3**	**2**	**3**	**4**	**1**	**3**	**3**	**2**	**2**
Question	16	17	18	19	20										
Answer	**3**	**2**	**2**	**1**	**3**										

Exercise - II — Electrochemistry

1. The equilibrium constant of the reaction :
$Zn(s) + 2Ag^+(aq) \rightarrow Zn^{2+}(aq) + 2Ag(s)$;
$E° = 1.50V$ at 298 K is
(1) 2.6×10^{49} (2) 8.7×10^{51}
(3) 6.1×10^{30} (4) 6.6×10^{50}

CEL220

2. The equivalent conductance of $\frac{M}{20}$ solution of a weak monobasic acid is 10 S cm^2/eq. and at infinite dilution is 200 S cm^2/eq. The dissociation constant of this acid is :-
(1) 1.25×10^{-4} (2) 1.25×10^{-5}
(3) 1.25×10^{-6} (4) 6.26×10^{-4}

CEL221

3. Which of the following expression correctly represents the equivalent conductance at infinite dilution of $Ca_3(PO_4)_2$ Given $\Lambda^\circ_{Ca^{+2}}$ and $\Lambda^\circ_{PO_4^{-3}}$ are the equivalent conductance at infinite dilution of the respective ions :-
(1) $\Lambda^\circ_{Ca^{+2}} + \Lambda^\circ_{PO_4^{-3}}$ (2) $\left(\Lambda^\circ_{Ca^{+2}} + \Lambda^\circ_{PO_4^{-3}}\right) \times 6$
(3) $\frac{1}{2}\Lambda^\circ_{Ca^{+2}} + \frac{1}{3}\Lambda^\circ_{PO_4^{-3}}$ (4) $3\Lambda^\circ_{Ca^{+2}} + 2\Lambda^\circ_{PO_4^{-3}}$

CEL222

4. Limiting molar conductivity of CH_3COOH (i.e. $\Lambda^\circ_{m(CH_3COOH)}$) is equal to :-
(1) $\Lambda^\circ_{m(CH_3COOH)} + \Lambda^\circ_{m(CH_3COONa)} - \Lambda^\circ_{m(NaOH)}$
(2) $\Lambda^\circ_{m(CH_3COONa)} + \Lambda^\circ_{m(HCl)} - \Lambda^\circ_{m(NaCl)}$
(3) $\Lambda^\circ_{m(CH_3COONa)} + \Lambda^\circ_{m(NaCl)} - \Lambda^\circ_{m(NaOH)}$
(4) $\Lambda^\circ_{m(NaOH)} + \Lambda^\circ_{m(NaCl)} - \Lambda^\circ_{m(CH_3COONa)}$

CEL223

5. The resistance of conductivity cell containing 0.001 M KCl solution at 298 K is 1500 ohm. What is the cell constant if the conductivity of 0.001 M KCl solution at 298 K is 0.146×10^{-3} S cm^{-1}
(1) 0.419 cm^{-1} (2) 0.219 cm^{-1}
(3) 0.45 cm^{-1} (4) 0.75 cm^{-1}

CEL224

6. A solution of $CuSO_4$ is electrolysed for 10 minutes with a current of 1.5 amperes. What is the mass of copper deposited at the cathode ?
(Molar mass of Cu = 63.5 g/mol)
(1) 0.492g (2) 0.325g
(3) 0.873g (4) 0.296g

CEL225

7. The molar conductivity of 0.01 M solution of weak acid is 16.6 Ω^{-1} cm^2 mol^{-1}. Its molar conductivity at infinite dilution is 390.7 Ω^{-1} cm^2 mol^{-1}. The degree of dissociation of weak acid is
(1) 0.24 (2) 0.42 (3) 0.042 (4) 0.024

CEL226

8. In an electrolysis of acidulated water, 4.48 L of hydrogen at STP was produced by passing a current of 2.14 A. For how many hours was the current passed?
(1) 4 (2) 3 (3) 6 (4) 5

CEL227

9. The amount of metal deposited at cathode on passing electric current of 0.75 ampere in aqueous ferric sulphate solution for 30 minutes, will be (atomic weight of Fe = 56)
(1) 0.00435 g (2) 0.261 g
(3) 0.783 g (4) 0.522g

CEL228

10. The quantity of electricity required to release 112 cm^3 of hydrogen at STP from acidified water is
(1) 0.1F (2) 1F
(3) 965C (4) 96500C

CEL229

Exercise-II — ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	**4**	**1**	**1**	**2**	**2**	**4**	**3**	**4**	**2**	**3**

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Exercise - I Solutions

1. In an aqueous solution ethylene glycol has the mass percentage (% w/W) 30% then the mole fraction of ethylene glycol will be :-
(1) 0.428 (2) 0.124
(3) 0.11 (4) 0.889
CSO232

2. What will be the molality of solution prepared by dissolving 3.7 g propanoic acid in 80 g benzene ?
(1) 0.77 m (2) 0.625 m
(3) 0.045 m (4) 46.25 m
CSO233

3. When HgI_2 is added in KI solution. The freezing point of solution :-
(1) Increases
(2) Decreases
(3) Remains unchanged
(4) Can't predict
CSO234

4. A solute dissociates in solution according to reaction $2A \rightarrow 5B$, If solute shows 30% dissociation then van't Haff factor will be :-
(1) 2.2 (2) 1.45 (3) 2.9 (4) 1.9
CSO235

5. Which of the following relationship is correct ?
(1) $K_b = \frac{RT_b^2}{1000 \times \Delta H_v}$ (2) $K_b = \frac{RT_b^2}{1000\, L_v}$
(3) $K_f = \frac{RT_f^2}{1000 \Delta H_f}$ (4) $K_f = \frac{RT_f^2 \times M}{1000 \times L_f}$
CSO236

6. Which of the following relationship is correct?
(A $\rightarrow$ solvent; B $\rightarrow$ solute)
(1) $\frac{P^o - P_S}{P^o} = X_A$ (2) $Y_B = \frac{P_B^o - X_B}{P_A^o}$
(3) $P_A = P_A^o X_A$ (4) $P_S = P_A^o X_A + P_B^o X_A$
CSO237

7. Van't Hoff factor is 1.92 for MgI_2 solution with concentration 0.2M then the degree of dissociation of salt at this concentration is :-
(1) 46% (2) 96%
(3) 30.67% (4) 64%
CSO238

8. 18 g glucose ($C_6H_{12}O_6$) is added to 178.2 g of H_2O the vapour pressure of water for this aqueous solution at 100°C.
(1) 759 torr (2) 7.60 torr
(3) 76 torr (4) 752.4 torr
CSO239

9. The solubility of gases in liquid is maximum at :-
(1) Low temperature and low pressure
(2) Low temperature and high pressure
(3) High temperature and high pressure
(4) High temperature and low pressure
CSO240

10. What will be the osmotic pressure of 0.03 N Aluminium sulphate solution at 27°C? If in solution salt dissociates 90% :-
(1) 0.566 atm (2) 0.677 atm
(3) 3.399 atm (4) 4.065 atm
CSO241

11. Which of the following will have maximum depression in freezing point?
(1) 0.5 M Li_2SO_4 (2) 1M KCl
(3) 0.5 M $BaCl_2$ (4) 1M $Al_2(SO_4)_3$
CSO242

12. What will be normality of "20V H_2O_2" solution?
(1) 1.78 N (2) 3.57 N
(3) 10N (4) 0.28 N
CSO243

13. The van't Hoff factor for 0.1M $Ba(NO_3)_2$ solution is 2.74. The degree of dissociation is
(1) 91.3% (2) 87%
(3) 100% (4) 74%
CSO244

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14. In which case van't Hoff factor is maximum?
(1) KCl, 50% ionised
(2) K_2SO_4 40% ionised
(3) $FeCl_3$,30% ionised
(4) $SnCl_4$, 20% ionised

CSO245

15. If liquid A and B form ideal solution, then :-
(1) $\Delta G_{mix} = 0$
(2) $\Delta H_{mixing} = 0$
(3) $\Delta G_{mix} = 0$, $\Delta S_{mix} = 0$
(4) $\Delta S_{mix} = 0$

CSO246

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Exercise-I ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	**3**	**2**	**1**	**2**	**2**	**3**	**1**	**4**	**2**	**1**	**4**	**2**	**2**	**3**	**2**

Exercise - II — Solutions

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1. 10% (w/V) solution of glucose is isotonic with 4% (w/V) solution of non-volatile solute then molar mass of non-volatile solute will be :-

(1) 36 g mol^{-1} (2) 72 g mol^{-1}

(3) 54 g mol^{-1} (4) 63 g mol^{-1}

CSO247

2. The vapour pressure of mixture of toluene and xylene at 90°C is 0.5 atm. If at this temperature vapour pressure of pure toluene and xylene are 400 mm Hg and 150 mm Hg respectively then what will be the mole fraction of toluene in mixture?

(1) 0.08 (2) 0.92

(3) 0.88 (4) 0.46

CSO248

3. What will be the freezing point of 0.2 molal aqueous solution of $MgBr_2$? If salt dissociates 40% in solution and K_f for water is 1.86 K kg mol^{-1}

(1) –3.35°C (2) –0.67°C

(3) –0.6°C (4) –0.45°C

CSO249

4. Which of the following solution has maximum vapour pressure :-

(1) 1 N KNO_3

(2) 1 N $Ba(NO_3)_2$

(3) 1 N $Al_2(SO_4)_3$

(4) 1 N $Ti(NO_3)_4$

CSO250

5. The vapour pressure of pure liquid A and liquid B at 350 K are 440 mm and 720 mm of Hg. If total vapour pressure of solution is 580 mm of Hg then the mole fraction of liquid A in vapour phase will be :-

(1) 0.31 (2) 0.38

(3) 0.62 (4) 0.76

CSO251

6. The boiling point of solution obtained by dissolving 0.51 g anthracene in 35 g chloroform increases by 0.32°C then what will be the molar mass of anthracene if for chloroform K_b = 3.9 K kg mol^{-1} :-

(1) 175.2 g mol^{-1} (2) 177.6 g mol^{-1}

(3) 178.6 g mol^{-1} (4) 182.3 g mol^{-1}

CSO252

7. Which of the following statement is **incorrect**:-

(1) In solution of volatile liquids, partial pressure of any component is directly proportional to its mole fraction in solution.

(2) Components of azeotropic mixture can be separated in pure form using fractional distillation

(3) Total vapour pressure of solution of binary volatile liquids, decreases on increasing mole fraction of less volatile component.

(4) At a given pressure, higher the value of K_H, the lower is the solubility of the gas in the liquid.

CSO253

8. 1.00 g of a non electrolyte solute is dissolved in 50g of benzene which lowers the freezing point of benzene by 0.40K. The freezing point depression constant of benzene is 5.12 K kg mol^{-1}. Find the molar mass of the solute

(1) 206 g mol^{-1} (2) 226 g mol^{-1}

(3) 246 g mol^{-1} (4) 256 g mol^{-1}

CSO254

9. A binary liquid solution is prepared by mixing n-heptane and ethanol. Which one of the following statement is correct regarding the behaviour of the solution?

(1) The solution formed is an ideal solution

(2) The solution is non-ideal, showing positive deviation from Raoult's law

(3) The solution is non ideal, showing negative deviation from Raoult's law

(4) n-heptane shows positive deviation while ethanol shows negative deviation from Raoult's law

CSO255

10. What is the correct sequence of osmotic pressure of 0.01 M aq solution of

(a) $Al_2(SO_4)_3$ (π_1)

(b) Na_3PO_4 (π_2)

(c) $BaCl_2$(π_3)

(d) Glucose (π_4)

Choose the correct option.

(1) $\pi_4 > \pi_2 > \pi_3 > \pi_1$

(2) $\pi_3 > \pi_4 > \pi_2 > \pi_1$

(3) $\pi_3 > \pi_4 > \pi_1 > \pi_2$

(4) $\pi_1 > \pi_2 > \pi_3 > \pi_4$

CSO256

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Exercise-II ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	**2**	**2**	**2**	**3**	**2**	**2**	**2**	**4**	**2**	**4**

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Exercise - I Chemical Kinetics

1. Which of the following rate law expression represent zero order reaction :-
(1) $K=[A]^{3/2}[B]^{-1}[C]^{1/2}$
(2) $K=[A]^{\circ}[B]^{2}$
(3) $K=[A]^{1/2}[B]^{1/2}[C]^{-1}$
(4) $K=[A]^{2}[B]^{1}$

CCK224

2. The correct expression for Arrhenius equation:-
(1) $\ell nK=\ell nA-\frac{E_a}{RT}$
(2) $Ke^{-E_a/RT}=A$
(3) $\log K=\log A-2.303\frac{E_a}{RT}$
(4) $K=A^{-1}e^{-E_a/RT}$

CCK225

3. For every 10°C increase in temperature, the rate becomes twice. If the temperature increases from 10°C to 100°C then rate increases .... times :-
(1) 400 (2) 512
(3) 112 (4) 614

CCK226

4. The rate constant for zero order reaction is 3×10^{-2} mol L^{-1} s^{-1}. After 25 sec, if the concentration of reactant is 0.5M then initial concentration of the reactant is :-
(1) 1 M (2) 1.25 M
(3) 0.5 M (4) 0.75

CCK227

5. In a reaction $N_{2(g)}+3H_{2(g)}\rightarrow 2NH_{3(g)}$ rate of appearance of NH_3 is 2.5×10^{-4} mol L^{-1} s^{-1} then rate of reaction and rate of disappearance of H_2 respectively is :- (in mol L^{-1} s^{-1})
(1) 3.75×10^{-4}, 1.25×10^{-4}
(2) 1.25×10^{-4}, 2.5×10^{-4}
(3) 1.25×10^{-4}, 3.75×10^{-4}
(4) 5×10^{-4}, 3.75×10^{-4}

CCK228

6. Calculate the order of the reaction with respect to A and B :-

A (mol/L)	B (mol/L)	Rate (mol L^{-1} s^{-1})
0.05	0.05	1.2×10^{-3}
0.10	0.05	2.4×10^{-3}
0.05	0.10	1.2×10^{-3}

(1) 1 and 0 (2) 1 and 1
(3) 0 and 1 (4) none of these

CCK229

7. For a reaction rate $= K[A]^{1/2}[B]^{2}$. If concentration of A and B are increased by factor of 4 and 2 respectively then rate is changed by the factor of :-
(1) 4 (2) 6
(3) 8 (4) None of these

CCK230

8. The rate constant for first order reaction whose half life is 480 sec. :-
(1) 1.44×10^{-3} s^{-1} (2) 1.44 s^{-1}
(3) 0.72×10^{-3} s^{-1} (4) 2.88×10^{-3} s^{-1}

CCK231

9. The half-life of a first order reaction is 6 hour. How long will it take for the concentration of reactant to change from 0.8 M to 0.25 M?
(1) 1.07 hour (2) 5.1 hour
(3) 2.7 hour (4) 10.06 hour

CCK232

10. Which of the following is the correct expression for Arrhenius equation?
(1) $\ln\frac{K_2}{K_1}=\frac{E_a}{R}\left(\frac{1}{T_1}-\frac{1}{T_2}\right)$
(2) $\log\frac{K_2}{K_1}=\frac{E_a}{2.303}\left(\frac{T_1T_2}{T_2-T_1}\right)$
(3) $\ln\frac{K_2}{K_1}=\frac{E_a}{2.303R}\left(\frac{1}{T_1}-\frac{1}{T_2}\right)$
(4) $\frac{K_2}{K_1}=\frac{E_a}{2.303R}\left(\frac{T_2-T_1}{T_1T_2}\right)$

CCK233

11. For the reaction, $2N_2O_5 \rightarrow 4NO_2 + O_2$ rate and rate constant are 1.02×10^{-4} M s^{-1} and 3.4×10^{-5} s^{-1} respectively, the concentration of N_2O_5, at that time will be

(1) 1.732 M (2) 3M

(3) 1.02×10^{-4} M (4) 3.5×10^{5} M

CCK234

12. For the reaction $N_2O_5(g) \rightarrow N_2O_4(g) + ½O_2(g)$ initial pressure is 114 mmHg and after 20 seconds the pressure of reaction mixture becomes 133 mmHg then the average rate of reaction will be

(1) 1.9 atm s^{-1}

(2) 8.75×10^{-3} atm s^{-1}

(3) 2.5×10^{-3} atm s^{-1}

(4) 6.65 atm s^{-1}

CCK235

13. The rate of reaction is expressed

$$+\frac{1}{2}\frac{d[C]}{dt} = -\frac{1}{3}\frac{d[D]}{dt} = +\frac{1}{4}\frac{d[A]}{dt} = \frac{-d[B]}{dt}$$

the reaction is

(1) $4A + B \rightarrow 2C + 3D$ (2) $B + 3D \rightarrow 4A + 2C$

(3) $A + B \rightarrow C + D$ (4) $B + D \rightarrow A + C$

CCK236

14. Rate constant for first order reaction is 5.78×10^{-5} s^{-1}. What percentage of initial reactant will react in 10 hours?

(1) 12.5% (2) 25% (3) 87.5% (4) 75%

CCK237

15. For a first order reaction -

(1) The degree of dissociation is equal to e^{-kt}

(2) A plot of reciprocal concentration of the reactant vs time gives a straight line

(3) The time taken for the completion of 75% reaction is thrice the $t_{1/2}$ of the reaction

(4) The pre-exponential factor in the Arrhenius equation has the dimension of time, T^{-1}

CCK238

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Exercise-I ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	3	1	2	2	3	1	3	1	4	1	2	3	2	3	4

Exercise - II
Chemical Kinetics

1. The ratio of $t_{0.875}$ and $t_{0.75}$ for first order reaction:-
(1) 4 : 3 (2) 3 : 2 (3) 2 : 1 (4) 1 : 2
CCK239

2. For a first order gaseous reaction $A \rightarrow 2B + C$, initial pressure is P_i and total pressure after time 't' is P_t then expression of rate constant:-
(1) $K = \frac{2.303}{t}\log\frac{2P_i}{3P_i - P_t}$
(2) $K = \frac{2.303}{t}\log\left(\frac{2P_i}{2P_t - P_i}\right)$
(3) $K = \frac{2.303}{t}\log\left(\frac{P_i}{P_i - P_t}\right)$
(4) None of these
CCK240

3. The rate law of reaction A + 2B $\rightarrow$ product is given by
$r = K[A]^2[B]^1$. If A is taken in excess the order of the reaction will be :-
(1) zero (2) 1 (3) 2 (4) 3
CCK241

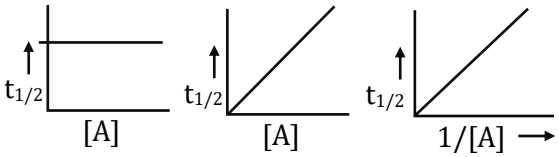
4. The half-life for the first order reaction $A \longrightarrow B + C$ is 24 hrs. starting with 20g of A how many grams of A will remain after 72 hours.
(1) 1.25g (2) 0.63g (3) 1.77g (4) 2.5g
CCK242

5. K is rate constant at temp T then value of $\lim_{T\to\infty} \log K$ is equal to :-
(1) A/2.303 (2) A
(3) 2.303 A (4) log A
CCK243

6. The reaction 2A + B $\rightarrow$ product follows the mechanism :-
$2A \rightleftharpoons A_2$(fast) $A_2 + B \rightarrow P$(slow)
The order of the reaction is
(1) 1.5 (2) 3 (3) 1 (4) 2
CCK244

7. Which of following represents the expression for $\frac{3}{4}$th life of first order reaction :-
(1) $\frac{K}{2.303}\log\frac{4}{3}$ (2) $\frac{2.303}{K}\log\frac{3}{4}$
(3) $\frac{2.303}{K}\log 4$ (4) $\frac{2.303}{K}\log 3$
CCK245

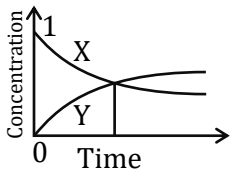
8. consider the plots for a reaction $nA \rightarrow B + C$
These plots respectively correspond to the reaction order



(I) (II) (III)

(1) 0,1,2 (2) 1,2,3
(3) 1,0,2 (4) None of these
CCK246

9. The accompanying figure depicts the change in concentration of species X and Y for the reaction $X \rightarrow 3Y$ as a function of time the point of intersection of the two curves represents.



(1) $t_{1/2}$
(2) $t_{3/4}$
(3) $t_{1/4}$
(4) Data insufficient to predict
CCK247

10. In the following reaction : $xA \longrightarrow yB$
$$\log\left[-\frac{d[A]}{dt}\right] = \log\left[\frac{d[B]}{dt}\right] + 0.3010$$
where –ve sign indicates rate of disappearance of the reactant. Thus, x : y is :
(1) 1 : 2 (2) 2 : 1 (3) 3 : 1 (4) 3 : 10
CCK248

Exercise-II
ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	2	1	2	4	4	2	3	3	3	2

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